

25 - 27 september — 2026

Solemtrix Vineyard AI

For vineyard inspectors and farmers who can't check every vine on foot, we built an offline AI pipeline and web map that turns one drone flight into a per-plant inventory and a 24 km inspection walk.

CHALLENGE

AI/ML · AgriFood · GeoTech — Vineyard AI Field Challenge
(Siret3) — Marcaj

TEAM

Solemtrix Hardware & Software — Miroslav
Maletici, Razvan Pervulescu, [other members]

The problem

WHO HAS THIS PROBLEM?

State vineyard inspectors (vine registry, subsidy checks), farmers and agronomists who need to know every row and every plant.

WHAT HAPPENS TODAY?

An inspector walks every inter-row: 68 km and ≈ 17 h for the 81.5 ha of Siret³, and still has no per-plant record of gaps, missing vines or waste.

WHY DOES IT MATTER TO THE CHALLENGE PROVIDER?

Aerial images alone give no measured areas and no efficient walking plan, while the registry (45,500 ha) and statistics (66,400 ha) differ by 20,900 ha.

17 h

on foot to inspect one 81.5 ha
village: a 68 km serpentine
through every inter-row

Our solution

One drone flight in: every vineyard block, row and plant measured with an ID, checked in Marcaj, and the shortest closed inspection walk out. Offline, on one laptop, in ≈ 17 minutes.

1

Per-plant map

Click any block, row, plant or waste spot and see its ID and attributes: 674 rows, 13,702 plants.

2

Measurements

Counts, row lengths, canopy and inter-row areas per block and farm, tied to cadastre parcels.

3

Inspection route

Walk, or send a robot, along a closed 24 km route through 795 targets; GPX per farm.

Demo



SCREENCAST

[YouTube link]

ONE USER, ONE TASK

An inspector opens farm F09, block V13, row V13-R020: counts, areas, gaps, then the route as GPX.

PRE-TEST POINT

47.1225 N, 28.7095 E: block V13, inter-row V13-I020; our route starts and ends here (0.00 m).

TOP-5 additional features

1

Field robot

A working ESP32 ground robot driven from the web app: live camera with pan / tilt, panoramas in one press, hold-to-drive and a 20 cm obstacle stop.

2

Grape and disease detector

Our own YOLO (YOLO11 / 26) fine-tuned on public vineyard sets: grape bunches (WGISD, ViViD-5K), trunks, and diseased leaves (Flavescence dorée).

3

Cadastre and farm registry

Every block linked to its official AGCC cadastre parcels (codes, land use); blocks grouped into 27 farms; public vs internal roads. For the vine registry and AIPA subsidy checks.

4

Inspector team and tasks

Accounts per municipality: admin, inspectors, viewers. The admin assigns a spot or a whole farm to an inspector, who gets a notification; finished tasks are tracked.

5

In-app AI assistant

Answers *how do I...?* for each role in RO / EN / RU, quoting the exact button labels on screen (Gemini, keys kept on the server).

How it works

INPUT

311 drone GeoTIFFs (2.5 cm/px, 81.5 ha, EPSG:32635), AGCC cadastre, OSM roads. Only 2 labelled tiles.



OUR ENGINE

CV first: rows as a 1-D signal, one canopy per plant, rule-based attributes; U-Net on 88 tiles; waste cascade; OR-Tools route.



OUTPUT

Marcaj annotations, measurements.csv, route GeoJSON / GPX, a web map with every ID (RO / EN / RU).

WHAT'S REAL

The full flow on all 311 real tiles in ≈ 17 min on one laptop CPU, no paid API. On the 2 labelled tiles: row F1 1.00, canopy 0.876, inter-row IoU 0.96.

WHAT'S STILL MOCKED

A human still checks every tile in Marcaj. The robot is a prototype; the grape and disease detectors are trained on public datasets, not yet validated on Siret3. Registry / AIPA export is not built. Deployment hours are estimates.

Impact

-65 %

shorter inspection walk: 24.2 km
instead of a 68.2 km serpentine

11.0 h

inspector time saved per
survey: 6.0 h instead of 17 h

17 min

from 311 tiles to deliverables on
one laptop CPU, no paid API

WHAT THIS MEANS FOR THE CHALLENGE PROVIDER

Marcaj's users get measured areas and an efficient walk, not just pictures. Deployment takes $\approx$ 11–20 h from zero to the first inspected region: 6–11 h one-time setup, then 5–9 h per survey, mostly the human check. Route and compute are measured on Siret3 at 4 km/h; setup hours are estimates.

Next steps



NEXT WEEK

Run a second village from the next drone flight; export results per cadastre parcel for the vine registry.



NEXT MONTH

Season-to-season change per row; an offline Docker image for ministry servers; per-farm GPX and CSV reports.



IN 3 MONTHS

A ground robot drives our route: side cameras, on-board disease detection, every image tied to its row_id.

WHAT WE NEED · HOW IT SCALES

More flights and 2 labelled tiles per region, and a pilot with the vine registry (ONVV) and AIPA. Measured 4.7 s/ha: all $\approx 100,000$ ha of Moldovan vineyards in ≈ 130 h on one laptop, ≈ 16 h on a 64-core server (extrapolated).

Team

MM

**Miroslav
Maletici**

AI & pipeline

CV pipeline, U-Net, route
solver, Marcaj export

RP

**Razvan
Pervulescu**

Web platform

Map, statistics, farm
routes, robot views

?

[Name]

[Role]

[One line: why you]

Team lead: [name]. Solemtrix Hardware & Software.

One drone flight: every vine measured with an ID, and a 24 km walk instead of 68 km. Offline, on one laptop, in 17 minutes.

Thank you!

Solemtrix Hardware & Software · [\[email / LinkedIn\]](#)
[github.com/miroslav43/Solemtrix_Hardware_Software__](https://github.com/miroslav43/Solemtrix_Hardware_Software_Vineyard_AI_Field_Challenge)
[Vineyard_AI_Field_Challenge](#)



Backup slides

Not part of the 5 minutes. Show them only if the jury asks.

Backup: details for Q&A

TECHNICAL DETAILS

Rows: dominant angle, 1-D projection, autocorrelation spacing (1.8–3.8 m), $\text{SNR} \geq 3$ rejects orchards, Huber fit, union-find across tiles.

Canopies: Lab a^* threshold inside the row corridor, one polygon per plant; U-Net only on 88 tiles (13 s).

Waste: colour and shape rules, OpenCLIP probe, SAM3 on the top crops; a human approves every box.

Route: walking graph on inter-rows and authorised passages, generalised TSP (OR-Tools, 2-opt), closed at START, 1.29 % outside.

2 labelled tiles: row F1 1.00 (51/51), canopy 0.876, inter-row IoU 0.96, counts 0.973.

Engineering: ≈ 17 min on an Apple M3 Pro CPU; 2,378 tests, CI, offline Docker.

BUSINESS / SUSTAINABILITY

Who would pay: the vine registry and AIPA per surveyed hectare for subsidy checks; farms and cooperatives per survey for replanting and clean-up maps; a robot service per inspected row.

Cost per survey: no paid API, no GPU, open weights. What remains is the drone flight and 4–8 h of human checking in Marcaj.

Keeps running: one config file, a pinned uv.lock and an offline Docker image (`--network none`) on government servers; the web app in RO / EN / RU.

Backup: the organisers' pre-test point



X 629663.8 · Y 5220195.3

EPSG:32635 = 47.1225039 N,
28.7094577 E

Tile siret3_r020_c013

Farm / block F09 / V13 (14 parcels)

**Object inter-row V13-I020, mixed
Rows V13-R020 at 0.82 m (disrupted,
228.94 m, 70 plants), V13-R019 at
1.43 m**

**Route starts and ends here: 0.00 m
closure (rule: 5 m)**

**Block V13 77 rows, 13,551 m, canopy
3,472 m², inter-row 28,732 m²**

Backup: deployment time, 11–20 h

ONE-TIME SETUP · 6–11 H

Server or laptop: Docker or <code>uv sync</code> , <code>pnpm install</code> , model weights from the GitHub release	1–2 h
Web app for inspectors: build, hosting, TLS, logins	2–3 h
Calibrate a new region: label 2 reference tiles, check the thresholds	2–4 h
Train the operators (inspector, agronomist)	1–2 h

PER NEW SURVEY (≈ 80 HA) · 5–9 H

Ingest and AI pipeline (measured: 2.5 s + ≈ 17 min)	0.5 h
Human check in Marcaj	4–8 h
Final run: route, measurements, reports, publish	0.5 h

Compute times are measured (README §6). Setup and human-check times are team estimates.